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Mind the fake reviews! Protecting consumers from deception through persuasion knowledge acquisition

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A R T I C L E I N F O

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A B S T R A C T

A growing body of research has shown that while computers can effectively detect fake reviews, humans are no more accurate than chance. Since consumers strongly trust online reviews, and fake reviews are pervasive, they often make suboptimal choices. However, whether consumers can learn to detect fake reviews and whether this knowledge would help them make better-informed decisions remain open questions. We propose that learning four distinctive features of fake reviews (one-sidedness, exaggeration, personal selling style, and generic descriptions) affects consumers' trustworthiness in them and their perceived favorability, thus affecting their purchase intentions toward the target product. Five studies support our theoretical model. We also show that one-sidedness is the most discriminating among the four features and that simply activating consumers' current knowledge is not enough to protect them from fake reviews.

1. Introduction

Online reviews have become a powerful instrument to persuade consumer decisions (Bastos and Moore, 2021; Cheung et al., 2008; Hornik et al., 2015; Rosario et al., 2016). For instance, recent market surveys have reported that over 90 % of consumers check online reviews before making purchasing decisions (BrightLocal, 2020; Kaemingk, 2020). This increasing reliance of consumers on online reviews can have a substantial impact on the performance of firms (Anderson and Magruder, 2012; Chevalier and Mayzlin, 2006; Dellarocas et al., 2007; Gössling et al., 2018). For example, research has revealed that a 1 % increase in hotel review ratings may increase sales per room by approximately 2.6 % (Gössling et al., 2018), and the likelihood of a restaurant's availability for a reservation drops from 90 % to 20 % when ratings move from 3.0 to 4.5 (Anderson and Magruder, 2012). Thus, it is paramount for managers to reach and maintain a high score level for their businesses or brands on the several internet platforms where consumers post their opinions.

By monitoring consumers' online reviews, managers can understand the strengths and shortcomings of their products and services, respond accordingly to capitalize on the strengths, and plan to correct deficiencies. However, many unscrupulous marketers and managers opt instead to engage in deception, artificially inflating the favorability of their products or services with positive fake reviews or attempting to damage the reputation of competitors with artificial negative reviews. Unfortunately, this is not just a practice conducted by a small niche of unprincipled marketers; fake reviews have become a big business. It is estimated that the prevalence of fake reviews varies from 16 % to more than 50 %, depending on the category and website (Camilleri, 2019).

Although online reviews have become an important source of information for consumers to make better-informed choices, the increasing prevalence of review manipulation potentially creates a bias in consumers toward suboptimal purchasing decisions because consumers have not yet developed the skills to cope with this menace. Consumers' levels of trust in online consumer reviews remain at staggeringly high levels (e.g., between 76 % and 84 % from 2015 to 2020, BrightLocal, 2020) and significantly more than in other traditional sources, such as advertising (Kantar, 2020). These facts suggest that consumers have not acquired the knowledge about persuasion attempts in online review contexts that they otherwise apply to other market-

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related contexts, such as advertising and personal selling (Munzel, 2016).

In the last decade, there has been prolific research on algorithms to detect fake reviews (e.g., Anderson and Simester, 2014; Luca and Zervas, 2016; Martinez-Torres and Toral, 2019; Mayzlin et al., 2014; Moon et al., 2021; Ong et al., 2014). This body of research has shown that computers are reasonably good at detecting fake reviews, whereas humans are generally no more accurate than chance (Hovy, 2016; Ott et al., 2011; Plotkina et al., 2020; Salminen et al., 2022), suggesting that either fake reviews are impossible to be detected by the human eye or consumers lack the knowledge to identify them.

Thus, the fundamental question of this research is as follows: can consumers learn to detect fake reviews, and if so, will they adjust their judgments accordingly to make better-informed purchasing decisions? This question remains largely understudied. A few recent papers have investigated the effect of activated or measured persuasion knowledge (i.e., the knowledge developed by consumers about persuasion attempts [Friestad and Wright, 1994]) in the context of online reviews (Bambauer-Sachse and Mangold, 2013; Munzel, 2015, 2016; Plotkina et al., 2020; Zhuang et al., 2018). However, whether consumers can learn how to distinguish fake from authentic reviews and how this affects their attitudes remains unknown.

The present research fills this gap. Through five experimental studies, we show that, when consumers acquire persuasion knowledge, they can distinguish fake from genuine reviews and adjust their judgments accordingly. Our research offers important contributions. First, we extend recent studies by providing persuasion knowledge rather than activating it. Second, we provide further evidence that persuasion knowledge enables consumers to better distinguish guileful from honest approaches rather than invariably eliciting skepticism, as many past studies have assumed (Campbell and Kirmani, 2000; Kirmani and Zhu, 2007). Third, we show that the causal mechanisms for the effect of persuasion knowledge acquisition on purchase intentions are adjustments in the reviews' favorability and trustworthiness. Finally, we reveal linguistic features of review content that discriminate fake from genuine reviews.

1.1. Persuasion knowledge

Over time, repeated interactions with firms through numerous touch points in multiple channels allow consumers to learn how marketing agents persuade them, enabling them to better discern trustworthy persuasion attempts from manipulative ones. This is the basis of Friestad and Wright's (1994) persuasion knowledge model (PKM). They defined persuasion knowledge as "folk knowledge" about persuasion intentions that people build on the experience in the marketplace from their interactions with persuaders, as well as with other sources, such as word-of-mouth communications, advice from parents, etc. This knowledge is stored, and it contains information about the marketers' persuasion goals, the tactics they use to achieve these goals, and their intended psychological effects, effectiveness, and appropriateness (Isaac and Grayson, 2017). Owing to this knowledge, consumers develop beliefs about how marketers try to influence them in a persuasion attempt, such as an advertising message or salesperson approach (Campbell and Kirmani, 2000). The PKM proposes that market agents (e.g., advertisers and salespeople) and targets (e.g., consumers) interact in a persuasion episode. A persuasion episode comprises the interaction between the consumer's persuasion-coping behaviors and the agent's persuasion attempts. On the one hand, market agents use their knowledge of how to persuade a target, together with their knowledge of the topic and the target, to attempt to persuade their targets. On the other hand, consumers use their knowledge of persuasion tactics employed by agents, together with their knowledge of the topic and the agent, to deploy strategies to cope with persuasion attempts. The more persuasion knowledge a consumer has, the higher his or her ability to recognize, reflect upon, and evaluate the intentions of persuasion agents (Eisend and Tarrahi, 2022).

Persuasion knowledge has been generally associated with consumer skepticism, as there is a general assumption that its role is to help consumers defend themselves against persuasion attempts (Isaac and Grayson, 2017, 2020). However, Friestad and Wright (1994) did not affirm that more persuasion knowledge invariably leads to greater resistance to persuasion attempts. Instead, they argue that the primary capacity of persuasion knowledge is to aid consumers in better distinguishing misleading from fair persuasive attempts. Thus, consumers with higher persuasion knowledge are not only more likely to recognize a manipulative intention and protect themselves from deception but are also more likely to recognize that "some tactics are used when marketers accurately understand and respect what people want to know about a type of product" (Friestad and Wright, 1994, p. 13). Indeed, there is empirical evidence that responses to marketing persuasion stimuli that differ in credibility are more distinctive among people of higher persuasion knowledge (e.g., Hamby and Brinberg, 2018; Isaac and Grayson, 2017; 2020; Ku and Chen, 2020).

1.2. Consumer persuasion knowledge and fake reviews

Fake reviews are manipulative attempts to persuade consumers to develop more (un)favorable attitudes toward the target product or service and to increase (decrease) the likelihood of its purchase. In contrast, genuine reviews reflect the faithful opinions of consumers about their actual experiences with a product or service and help others make better-informed decisions (Hennig-Thurau et al., 2004; Sundaram et al., 1998; Yoo and Gretzel, 2009). The effectiveness of a fake review—the persuasion attempt—will depend on how consumers integrate topic, agent, and persuasion knowledge. Consumers may suspect that the review is fake if they know the brand or product well (topic knowledge) and realize that the review inflates it or if they are aware that the website/vendor is suspicious (agent knowledge). In addition, if the consumer has some knowledge of the characteristics of a fake review (persuasion knowledge), he or she will be able to spot the scam and disregard the message. Of course, the higher the level of the three types of knowledge the consumer has, the more he or she will be able to identify fraud.

Most research has examined the role of persuasion knowledge by activating the baseline knowledge consumers have about persuasion attempts. Typically, in these studies, participants are stimulated to access their baseline persuasion knowledge by a prime or cue about the market agent's ulterior motives (Isaac and Grayson, 2017; Campbell and Kirmani, 2000; Kirmani and Zhu, 2007). For instance, Campbell and Kirmani (2000) activated persuasion knowledge by displaying a flattery comment by a salesperson before or after the sale. The flattery made before (vs after) activated to a greater extent the participants' persuasion knowledge (e.g., 'salespeople flatter customers to push sales'), which increased the participants' likelihood of rating the salesperson as insincere, dishonest, and manipulative.

Such studies are embedded in contexts such as advertising and sales, in which some baseline consumer persuasion knowledge is taken for granted, assuming that participants have had experience with several persuasion episodes in these contexts. However, this assumption is less straightforward in some contexts, such as online reviews. As previously noted, the high levels of trust consumers place in online reviews (BrightLocal, 2020; Camilleri, 2019) suggest that people are overall unsuspicious of manipulated reviews. Thus, the deception-aware mindset that consumers adopt in many market-related contexts, such as advertising, is much less likely to be adopted in online reviews, as consumers "do not wear the same shields of protection" in the case of online reviews" (Munzel, 2016, p. 97). Therefore, in the context of online reviews, we argue that consumers need to acquire persuasion knowledge (i.e., to learn how spammers try to persuade them by posting deceptive opinions on the internet).

The general hypothesis of this study is that when consumers acquire persuasion knowledge in online reviews, they will improve their accuracy in differentiating a fake from a genuine review, adjust their judgments accordingly, and make better purchasing decisions. As previously mentioned, the more persuasion knowledge consumers have, the more they will be able to distinguish between deceptive and honest persuasion attempts. In addition, not only will they resist misleading persuasion agents but also reward truthful ones. In the context of online reviews, this means that consumers who are unaware of the existence and dissemination of fake reviews on the internet are unlikely to spot them and are more likely to indiscriminately take all reviews as genuine. Thus, it should be expected that these naive consumers perceive fake and genuine reviews as equally trustworthy (i.e., similar in “the degree of confidence in the communicator’s intent to communicate the assertions he considers most valid”; Hovland et al. (1953), p. 21).

Since writers of fake reviews have an incentive to ‘sell’ the product, positive fake reviews, compared to authentic ones, are more one-sided and use more emotional, gushy, and promotional language (Filieri, 2016; Harris, 2012; Li et al., 2011; Moon et al., 2021; Ott et al., 2011; Yoo and Gretzel, 2009). Also, fake reviews use more positive language, complimentary adjectives (Costa et al., 2019; Martens and Maalej, 2019), exclamation marks, and hyperbolic language (Banerjee, 2022; Banerjee and Chua, 2014, 2017; Moon et al. (2019); Ott et al., 2011) more often. Consequently, fake reviews tend to be associated with more extreme evaluations (Feng et al., 2012; Luca and Zervas, 2016; Mukherjee et al., 2013). Thus, it is reasonable to assume that fake review writers, compared to their genuine counterparts, tend to post comments associated with higher favorability towards the product, that is, an overall positive sentiment of the review content that promotes and endorses the product.

Because fake reviews tend to be more favorable than genuine reviews, naive consumers who are less able to distinguish a fake from a genuine review might take all reviews at face value (i.e., as genuine reviews). Therefore, they will tend to judge a fake review more favorably and ultimately display higher purchase intentions, compared to genuine review. In contrast, consumers who acquire knowledge of review manipulation will likely have a finer ability to detect a fake review, become skeptical, discount the positivity of the review, and consequently lower their purchase intentions towards the reviewed product. Thus, the differences in the expected outcomes of consumers who acquire persuasion knowledge vs those who do not suggest that the level of persuasion knowledge moderates the effect of review authenticity on trustworthiness, review favorability, and purchase intentions. We formally introduce the following hypotheses for positive reviews:

H1: The effect of review authenticity on review trustworthiness will be moderated by persuasion knowledge acquisition about online reviews. Consumers who (do *not*) acquire persuasion knowledge will likely trust fake reviews less than genuine reviews (trust fake and genuine reviews equally).

H2: The effect of review authenticity on perceptions of review favorability will be moderated by persuasion knowledge acquisition about online reviews. Consumers who (do *not*) acquire persuasion knowledge will likely perceive positive fake reviews as less (more) favorable.

H3: The effect of review authenticity on purchase intentions will be moderated by persuasion knowledge acquisition about online reviews such that consumers who (do *not*) acquire persuasion knowledge will likely display lower (higher) purchase intentions for positive fake reviews when compared to genuine reviews.

The rationale above also suggests that trustworthiness and perceptions of review favorability mediate the influence of the authenticity of a review on consumers’ purchase intentions. First, reviews perceived as more (less) trustworthy should elicit higher (lower) intentions to purchase. Indeed, Zhuang et al. (2018) found support for the mediating role of suspicion (i.e., the flip side of trustworthiness) in purchase intentions. They revealed that manipulated reviews, compared to a control group of intact reviews, were trusted less by participants with high expertise (a proxy for persuasion knowledge), lowering their purchase intentions. On the other hand, for low-expertise participants, the path was null. We expect to find the same pattern of relationships. The mediating role of favorability should be straightforward, as perceptions of favorability is an attitudinal variable, and attitudes are correlated with behavioral intentions (Ajzen and Fishbein, 1980; Kim and Hunter, 1993). It should be expected that reviews perceived as more (less) favorable will elicit higher (lower) intentions to purchase.

We theorize that, after reading a review, consumers’ purchase intentions will be determined by the interplay of the trustworthiness of the review and perceptions of review favorability. According to our predictions, consumers who acquire persuasion knowledge will trust less and discount the favorability of a positive fake review relative to a genuine one and, as a result, will display lower purchase intentions. In contrast, consumers lacking this knowledge tend to exhibit greater purchase intention when exposed to a fake positive evaluation relative to a genuine one, as they perceive them as more favorable and equally trustworthy. Based on the above discussion, we formalize the following hypotheses:

H4: The interaction effect of review authenticity and persuasion knowledge on purchase intentions will be mediated by trustworthiness in the review such that fake and genuine reviews will be perceived as equally trustworthy among consumers who do *not* acquire knowledge, which will not affect purchase intentions, and genuine reviews will be perceived as more trustworthy than fake reviews among consumers who acquire persuasive knowledge, which will increase purchase intentions of the genuine vs the fake review.

H5: The interaction effect of review authenticity and persuasion knowledge on purchase intentions will be mediated by review favorability such that positive fake reviews will be perceived as more favorable than genuine reviews among consumers *without* persuasion knowledge. This will increase purchase intentions of the fake relative to the genuine review, while genuine reviews will be perceived as more favorable than positive fake reviews among consumers who acquire persuasion knowledge, which will increase purchase intentions of the genuine relative to the fake review.

Fig. 1 illustrates the theoretical framework of the hypotheses. Path a represents the moderated mediation by trustworthiness, while path b represents the moderated mediation by perceptions of review favorability.

2. Method

Our hypotheses were tested through five studies. We first performed a pilot study to examine consumers’ judgments after reading a positive fake vs a genuine review when no persuasion knowledge is acquired. As

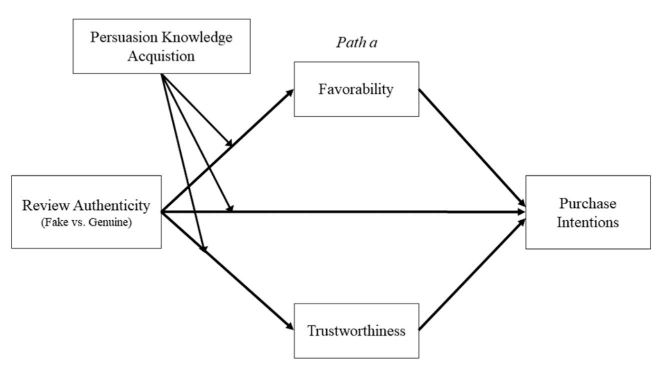

Fig. 1. Theoretical framework of the study.

expected, we found no differences in trustworthiness between the reviews, and the fake review was indeed judged as more favorable than the genuine review. Details of the pilot study are provided in Web Appendix G. In Study 1, we included a persuasion knowledge acquisition condition in which participants learned about the characteristics that distinguish fake reviews from genuine ones by reading a fake review detection guide. In Study 2, we added a third condition wherein the participant was only warned about the existence of fake reviews but did not learn how to spot them (i.e., persuasion knowledge activation). In Study 3, we tested our theoretical model for negative reviews.

In these studies, the participants were exposed to a review that was either genuine or fake. We created a fake review by altering parts of the text of a genuine review, adding features of fake reviews, or removing features typical of genuine reviews. We selected from the literature four characteristics that distinguish fake from genuine reviews: (1) balance and sidedness, (2) use of emotional language, (3) persuasive style, and (4) descriptions of the consumption experience. A summary of studies investigating features that distinguish fake from genuine reviews is shown in Web Appendix A. The process of selecting a true genuine review and the distinguishing features of fake vs genuine reviews are discussed in Web Appendix B.

Finally, we conducted follow-up studies to examine (a) the effect of our detection guide on consumers' accuracy in discerning a fake from a genuine review, (b) whether fake and genuine reviews are indeed perceived to contain more of the elements that characterize them, and (c) which of the features that distinguish fake from genuine reviews are more diagnostic.

In the pilot and first two studies, we measured consumer expertise with online reviews as a potential covariate, following Zhuang et al. (2018), who used expertise as a proxy for persuasion knowledge. Due to space limitations and because the results remained unchanged, we present the results of the covariate analyses in Web Appendix D. Likewise, we provide the descriptive statistics of the samples of all studies in Web Appendix C.

3. Study 1

The study's objective was to test our full theoretical model (H1 through H5).

3.1. Participants and design

Three hundred twenty-five Brazilian adults (73.0 % female; M_{age} = 41.0; SD = 13.2) participated in a 2 (persuasion knowledge: control vs acquisition) × 2 (review authenticity: fake vs genuine) between-subjects design online experiment.

3.2. Procedure

Participants were recruited through a paid Facebook advertisement in exchange for a raffle ticket to win a gift card. Results were not significantly different when we excluded the participants who failed attention check questions (N = 29), so our reports are based on the entire sample of 325 respondents (see Web Appendix D for results after exclusion).

Participants were randomly assigned to conditions. In the persuasion knowledge acquisition condition (henceforth "PK" condition), the participants read a paragraph containing a fake review warning and were then requested to read a fake review detecting guide that described the four characteristics that distinguished fake from genuine reviews (see Web Appendices A and B for details on the fake review characteristics and Web Appendix E for the detection guide). The participants assigned to the control condition were neither warned nor given any information about fake reviews. Next, the participants were asked to imagine browsing the internet for a restaurant for the weekend and finding a review. They then read either the genuine or the fake review of a restaurant (see the reviews in Web Appendix E).

3.3. Measures

After the manipulations, we measured participants' purchase intentions using items adapted from Munzel (2015) on a 9-point bipolar scale (1 = Low; 9 = High), the review's favorability on a 9-point semantic differential scale using four attitudinal items, and trustworthiness of the review on three items adapted from Munzel (2015) on a 9-point Likert-type scale (1 = Totally Disagree; 9 = Totally Agree). The trustworthiness of the review was measured in the end to avoid priming participants that the review could be fake.

Next, participants completed measures of their experience with shopping online and use of online reviews (1 = Totally Disagree; 9 = Totally Agree). The items were averaged to form a composite measure of expertise, which was tested as a potential covariate. The complete scales and reliabilities are provided in Web Appendix F. Finally, the participants completed demographic measures and were thanked for their participation. This procedure was pre-registered (https://aspredicted.org/blind.php?x=NY8_76S).

3.4. Results

We performed 2 × 2 ANOVAs on trustworthiness of the review, perceptions of favorability of the review, and purchase intentions as dependent variables, with persuasion knowledge acquisition and review authenticity as the independent variables. The results for each dependent variable are summarized in Tables 1 and 2.

3.4.1. Trustworthiness

As shown in Table 1, we observed a significant main effect for persuasion knowledge acquisition, as well as for review authenticity. Most importantly, the interaction effect of persuasion knowledge and review authenticity was also significant, as predicted in H1. As shown in Table 2, post hoc pairwise comparison tests by persuasion knowledge acquisition condition revealed no differences in trustworthiness in the reviews in the control condition but higher trustworthiness in the genuine review than in the fake one in the PK condition, lending support to H1a and H1b.

Table 1
ANOVA results (Study 1).

		F	η^2	M (SD)
(A) Persuasion Knowledge	Trustworthiness	12.8***	0.038	$M_{control}$ = 6.56 (1.94); M_{PK} = 5.69 (2.26)
	Favorability of the review	19.2***	0.056	$M_{control}$ = 7.43 (1.93); M_{PK} = 6.46 (2.07)
	Purchase Intentions	15.9***	0.047	$M_{control}$ = 7.16 (2.01); M_{PK} = 6.22 (2.23)
(B) Review Authenticity	Trustworthiness	10.1***.	0.030	M_{fake} = 5.75 (2.26); $M_{genuine}$ = 6.52 (1.96)
	Favorability of the review	2.54$^{n.s}$.	0.010	M_{fake} = 7.08 (2.09); $M_{genuine}$ = 6.79 (2.02)
	Purchase Intentions	1.46$^{n.s}$.	0.01	M_{fake} = 6.79 (2.17); $M_{genuine}$ = 6.57 (2.16)
(A) * (B)	Trustworthiness	10.6***.	0.032	
	Favorability of the review	10.3***.	0.031	
	Purchase Intentions	15.1***.	0.045	

Abbreviations: M = mean; SD = standard deviation; η^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

Table 2
Post hoc pairwise comparisons by persuasion knowledge acquisition conditions.

	PK				Control			
	Genuine	Fake	F	η^2	Genuine	Fake	F	η^2
Trustworthiness	6.48 (1.99)	5.02 (2.27)	20.9***	0.061	6.55 (1.95)	6.57 (1.95)	$0.01^{n.s}$	< 0.01
Favorability of the review	6.65 (2.10)	6.30 (1.78)	$1.33^{n.s}$	< 0.01	6.91 (1.94)	7.96 (1.78)	11.4***	0.034
Purchase Intentions	6.56 (2.17)	5.94 (2.25)	3.62*	0.011	6.58 (2.18)	7.76 (1.63)	12.8***	0.038

Abbreviations: M = mean; SD = standard deviation; h^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

3.4.2. Favorability of the review

As shown in Table 1, there was a significant main effect for persuasion knowledge, a non-significant effect for review authenticity, and most importantly, a significant persuasion knowledge × review authenticity interaction effect, as predicted in H2. Post hoc comparisons revealed that, while participants evaluated the fake review more favorably than the genuine review in the control condition, there were no significant differences between fake and genuine reviews in the PK condition (see Table 2). Therefore, H2a was supported but not H2b.

3.4.3. Purchase intentions

We observed a significant main effect for persuasion knowledge, a non-significant main effect of review authenticity, and, most importantly, a significant interaction effect of persuasion knowledge and authenticity of the review, as predicted in H3 (see Table 1). Further analysis revealed that purchase intentions were higher for the fake review than for the genuine review in the control condition. Conversely, in the PK condition, the participants displayed marginally higher purchase intentions (p =.06) after reading the genuine review compared to the fake one (see Table 2). Therefore, both H3a and H3b were supported.

3.4.4. Mediation analyses

A regression analysis using PROCESS macro model 8 (Hayes, 2018) with 10,000 bootstrap samples was conducted to test the mediation effects of trustworthiness and review favorability on purchase intentions. In the model, review authenticity (fake = 0; genuine = 1) was entered as the predictor, perceptions of the favorability of the review and trustworthiness as the mediators, persuasion knowledge acquisition (control = 0; acquisition = 1) as the moderator, and purchase intentions as the dependent variable. The direct effect of review authenticity on purchase intentions was significant in the control condition (b = -0.46, SE = 0.23, p =.04), but not in the PK acquisition condition (b = 0.19, SE = 0.23, p =.42). As predicted in H4, there was a significant moderated mediation effect by trustworthiness (index of moderated mediation: b = 0.19 [0.03: 0.42]). Consistent with our predictions, we found a positive and significant indirect effect of trustworthiness in the PK acquisition condition (0.19 [0.03: 0.42]) but not in the control condition (-0.002 [-0.09:0.09]). Therefore, H4 was fully supported. We also found a moderated mediation by review favorability (index: b = 0.95 [0.34: 1.60]), as predicted in H5. The conditional indirect effect of review authenticity on purchase intentions through review favorability was negative and significant in the control condition (-0.71 [-1.16:-0.30]) but not in the PK acquisition condition (0.24 [-0.19:0.67]). Therefore, H5 was partially supported.

3.5. Discussion

Study 1 confirmed our predictions: naive consumers trust both fake and genuine reviews similarly, attributing higher levels of favorability and higher purchase intentions for fake vs genuine reviews. In contrast, consumers can better discern truthful and deceitful reviews and adjust their judgments accordingly when they acquire persuasion knowledge about online reviews. They trust less, discount the favorability, and display lower purchase intentions toward fake reviews compared to genuine reviews.

Furthermore, our results show that the effect of review authenticity on purchase intentions is governed by different mechanisms that depend on persuasion knowledge. When no knowledge is provided, the fake review is perceived as more favorable and equally trustworthy compared to the genuine review, which renders higher purchase intentions of the fake vs the genuine review. However, when knowledge about fake reviews is acquired, the fake review is perceived as favorable and less trustworthy than genuine reviews, rendering higher purchase intentions towards the genuine vs fake review.

4. Study 2

Our overall hypothesis is that consumers adjust their ratings, make better decisions, and become more accurate in detecting fake reviews when they learn about them. However, if we just made consumers aware of fake reviews like some past studies did (Bambauer-Sachse and Mangold, 2013; Ma and Lee, 2014; Munzel, 2016), could this activate pre-existing knowledge and help them make better decisions too?

Recent studies have shown that, in general, consumers become more vigilant and activate a deceptive-aware mindset when warned about fake reviews (Bambauer-Sachse and Mangold, 2013; Ma and Lee, 2014; Munzel, 2015, 2016). However, because their accuracy is low (Plotkina et al., 2020), they tend to indiscriminately become suspicious of reviews in general, regardless of authenticity. Thus, activating persuasion knowledge by only warning consumers about the prevalence of deceptive reviews without giving them tools to discern a fake from a truthful review is likely to render them more suspicious and discount the valence of reviews altogether, regardless of review authenticity.

> **H6**: Consumers whose persuasion knowledge is activated (i.e., only warned about fake reviews) but do not acquire persuasion knowledge, compared to consumers whose persuasion knowledge is *not* activated, will: (a) trust less both fake and genuine reviews, (b) attribute less favorability to both fake and genuine reviews, and (c) display lower intentions to purchase the reviewed product in both fake and genuine reviews.

Therefore, we conducted Study 2 with the objective of testing H6 (and further testing H1-H5).

4.1. Participants and design

Four hundred twenty-three individuals (M_{age} = 35.3; SD = 12.0; 68 % female) participated in a 3 (persuasion knowledge: control vs activation vs acquisition) × 2 (review authenticity: fake vs genuine) between-subjects design online experiment. Participants were randomly assigned to the conditions.

4.2. Procedure

Participants were recruited from Prolific (N = 362), an international survey panel, in exchange for 1 GBP. The sample was complemented with participants from SurveyCircle (N = 61), a mutual support panel in which people earn credits for completing surveys. The procedure was the same as in Study 1, except that we included an additional condition—"PK activation"—wherein participants only received the warning part of the PK acquisition condition (see the stimuli in Web Appendix E). We used the same measures and scales as in Study 1 (see Web Appendix F). The participants who failed attention checks were automatically taken to the end of the survey and did not complete the study. This procedure was pre-registered (https://aspredicted.org/WXG_FKB).

4.3. Results

We performed 3 × 2 ANOVAs on the dependent variables (trustworthiness, favorability, and purchase intentions), with persuasion knowledge and review authenticity as the independent variables. The results for each dependent variable are summarized in Tables 3 and 4.

4.3.1. Trustworthiness

As shown in Table 3, the main effects were significant for both persuasion knowledge acquisition and review authenticity. Post hoc tests revealed that trustworthiness toward the review was lower in the PK activation compared to the control condition ($p <.01$). This reduction in trustworthiness occurred in both reviews, but it was only statistically significant in the fake review (mean difference fake = -1.65, $p <.01$; mean difference genuine = -0.25, $p =.27$). Overall, the results support H6a.

As predicted in H1, the persuasion knowledge acquisition × review authenticity interaction was significant. As shown in Table 4, post hoc pairwise comparisons by PK condition revealed that the fake review was perceived as significantly less trustworthy than the genuine review in the control condition. In the PK acquisition condition, the genuine review was judged more trustworthy than the fake review. Thus, we found support for H1b but not for H1a. Interestingly, in the PK activation condition, the genuine review was perceived as more trustworthy than the fake review. The results of the control and PK activation conditions suggest that participants had some baseline persuasion knowledge, and priming consumers with a warning activated it.

Despite the rejection of H1a, the overall assumption of our research that persuasion knowledge acquisition improves consumers' abilities to discern a fake from a truthful review and aids them in making better decisions remains supported. This is evidenced by the considerable differences between the effect sizes of the PK acquisition condition relative to the control and the PK activation conditions. In addition, post hoc tests by review authenticity revealed that the fake review was significantly less trusted in the PK acquisition relative to both the PK activation and control conditions (*ps* < 0.01).

Table 3
ANOVA results (Study 2).

		F	η^2	M (SD)
(A) Persuasion Knowledge	Trustworthiness	22.8***	0.010	$M_{control}$ = 6.57 (1.55); $M_{PKactivation}$ = 5.60 (2.00); $M_{PKacquisition}$ = 5.21 (2.78)
	Review Favorability	13.4***	0.061	$M_{control}$ = 7.07 (1.75); $M_{PKactivation}$ = 6.43 (1.70); $M_{PKacquisition}$ = 6.08 (1.99)
	Purchase Intentions	10.4***	0.047	$M_{control}$ = 5.69 (2.30); $M_{PKactivation}$ = 4.71 (1.85); $M_{PKacquisition}$ = 5.00 (1.94)
(B) Review Authenticity	Trustworthiness	237***·	0.36	M_{fake} = 4.58 (2.18); $M_{genuine}$ = 7.02 (1.96)
	Review Favorability	16.1***	0.037	M_{fake} = 6.84 (2.16); $M_{genuine}$ = 6.22 (1.44)
	Purchase Intentions	9.7**.	0.023	M_{fake} = 5.42 (2.11); $M_{genuine}$ = 4.86 (2.02)
(A) * (B)	Trustworthiness	58.5**·	0.22	
	Review Favorability	26.3***	0.11	
	Purchase Intentions	20.4***	0.09	

Abbreviations: M = mean; SD = standard deviation; η^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

4.3.2. Review favorability

The results revealed significant main effects for both persuasion knowledge and review authenticity (see Table 3). Post hoc tests revealed that perceptions of review favorability were significantly lower in the PK activation compared to the control condition ($p <.01$). Furthermore, this reduction in favorability was observed for both reviews (mean difference fake = -0.77, $p <.01$; mean difference genuine = -0.61, $p <.01$). Thus, hypothesis H6b is supported.

As predicted in H2, the interaction of persuasion knowledge and review authenticity was also significant (see Table 3). Post hoc tests revealed that the fake review, compared to the genuine one, was judged as more favorable in the control condition but less favorable in the PK acquisition condition, supporting H2a and H2b. Interestingly, in the PK activation condition, the fake review was judged more favorably than the genuine review.

4.3.3. Purchase intentions

The main effects for both persuasion knowledge and review authenticity were significant. Post hoc tests revealed that purchase intentions were lower in the PK activation condition than in the control condition ($p <.01$) and for both reviews (mean difference fake = -1.24, $p <.01$; mean difference genuine = -0.81, $p <.01$), supporting H6c.

As predicted in H3, the interaction of persuasion knowledge and review authenticity was also significant (see Table 3). As shown in Table 4, pairwise comparisons revealed that, whereas purchase intentions were higher for fake reviews than genuine reviews in the control condition, the reverse occurred in the PK acquisition condition, supporting both H3a and H3b. Interestingly, in the PK activation condition, purchase intentions were higher for the fake review than for the genuine one.

4.3.4. Mediation analysis

A mediation analysis similar to that in Study 1 was conducted, except that persuasion knowledge was entered as a multicategorical variable in the model (W1: control = 0, PK activation = 1, PK acquisition = 0; W2: control = 0, PK activation = 0, PK acquisition = 1). The direct effect was not moderated by PK, so we used Hayes' (2018) model 7. There was a negative and significant direct effect (b = -0.94, SE = 0.20, $p <.01$). Indices of moderated mediation by trustworthiness were significant for both dummy coded variables (W1: 0.42 [0.17: 0.73]; W2: 1.22 [0.75: 1.72]). The mediating path through trustworthiness was positive and significant for all conditions (control: 0.18 [0.03:0.36]; PK activation: 0.60 [0.34:0.89]; PK acquisition: 1.40 [0.89:1.91]). Therefore, H4 is partially supported, as we did not predict mediation by trust in the control condition (H4a). The index of moderated mediation by review favorability was significant only for W2 (W1: 0.09 [-0.32: 0.48]; W2: 1.47 [0.97: 2.03]), meaning that there was no difference between the mediation effects between the control and the PK activation conditions. As in Study 2, the results revealed a negative and significant indirect effect of review authenticity on purchase intentions through review favorability in the control condition (-0.89 [-1.23:-0.58]). The results in the PK activation condition were very similar to those in the control condition (-0.79 [-1.17:-0.48]). In the PK acquisition condition, the

Table 4
Post hoc pairwise comparisons by persuasion knowledge acquisition conditions.

	PK Acquisition				PK Activation				Control			
	Genuine	Fake	F	η ²	Genuine	Fake	F	η ²	Genuine	Fake	F	η ²
Trustworthiness	7.65	3.01	295***	0.41	6.59	4.61	53.0***	0.11	6.84	6.26	4.7**	0.011
	(1.34)	(1.70)			(1.62)	(1.85)			(1.41)	(1.65)		
Review Favorability	6.61	5.59	12.8***	0.030	5.72	7.14	24.3***	0.055	6.33	7.91	31.6***	0.071
	(1.25)	(2.38)			(1.42)	(1.68)			(1.51)	(1.62)		
Purchase Intentions	5.57	4.48	11.2***	0.026	4.10	5.31	13.5***	0.031	4.91	6.55	25.8***	0.058
	(1.72)	(1.99)			(1.73)	(1.78)			(2.28)	(2.01)		

Abbreviations: M = mean; SD = standard deviation; h^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

effect was positive and significant (0.58 [0.22:0.96]). Thus, H5 was fully supported.

4.4. Discussion

Study 2 replicated most of the results of Study 1. When consumers learn about fake review characteristics, they adjust their judgments, discounting favorability and reducing their purchase intentions for fake reviews, compared to naive consumers. In contrast to Study 1, the participants in the control group attributed slightly lower trustworthiness to fake reviews, suggesting that they had some baseline persuasion knowledge. However, this effect substantially increased when consumers learned how to differentiate a fake review from a genuine one.

Furthermore, Study 2 examined the effect of persuasion knowledge activation (vs acquisition). We predicted that warning without learning would only activate a cautious mindset and increase general suspicion toward both fake and genuine reviews. Our assumptions were partially supported. Overall, the participants warned about fake reviews judged reviews less favorably, trusted them less, and displayed lower intentions to purchase the reviewed product of both reviews than those who did not. However, they did perceive the fake review as less trustworthy than the genuine review, suggesting that the prime activated some existing persuasion knowledge. Interestingly, though, the lower trustworthiness of fake reviews did not pull back their disposition to visit the restaurant. On the contrary, the fake review was judged more favorably and was associated with higher purchase intentions than the genuine review. In other words, warning participants made them more suspicious of a fake vs a genuine review, but curiously, this did not pull back their intentions to purchase the product from the fake review.

Most importantly, consistent with the overall hypothesis of the study, the positive differences in trustworthiness between genuine and fake reviews were considerably enlarged when consumers were taught, rather than just warned about fake reviews. This suggests that acquiring persuasion knowledge is much more effective in helping consumers discern fake from genuine reviews. More importantly, as opposed to participants who were just warned, those who also acquired knowledge did pull back their evaluations and purchase intentions toward the restaurant depicted in the fake review.

5. Study 3

In this study, we test our theoretical model also for negative reviews. We should expect similar results as those of the previous studies for trustworthiness and inverted patterns for favorability and purchase intentions because the negative impact of negative reviews on product evaluations should be weakened if consumers know that such reviews can be manipulated than in the case where consumers do not have this knowledge. We also use a new category: cleaning service.

5.1. Participants and design

Three hundred ninety-eight individuals (M_{age} = 39.8; SD = 14.3; 49.7 % female) participated in a 3 (persuasion knowledge: control vs activation vs acquisition) × 2 (review authenticity: fake vs genuine) between-subjects design online experiment. Participants were randomly assigned to conditions.

5.2. Procedure

Participants were recruited from Prolific in exchange for 1 GBP. The procedure was the same as in Study 2, except that we adapted the detection guide and the review stimuli for negative reviews (see Web Appendix E). In addition, in this study, we included a manipulation check at the end by having the participants answer the question, “In your opinion, the review you read is…” (1 = definitely fake; 9 = definitely authentic). We used the same measures and scales as in Study 2, with two main changes. We did not include expertise, as it did not have an effect in the previous studies (see Web Appendix F), and we included a single-item question to measure participants’ willingness to search for more information. We included this measure since participants may have low purchase intentions after reading a negative review; thus, they may be willing to search for more information to try to update their impression after suspecting that a review is fake. The participants who failed attention checks were automatically taken to the end of the survey and did not complete the study. This procedure was pre-registered (https://aspredicted.org/5BP_4FK).

5.3. Results

We performed the same 3 × 2 ANOVAs as in Study 2. The results are presented for each dependent variable and are summarized in Tables 5 and 6.

5.3.1. Manipulation check

There was a significant PK vs review authenticity interaction effect ($F(2, 392) = 26.7$, $p < 0.01$; $\eta^2 = 0.12$). The fake review was judged as significantly less authentic than the genuine review in the PK activation condition ($M_{fake} = 4.95$, $SD = 1.79$; $M_{genuine} = 6.22$, $SD = 1.62$; $F(1, 392) = 20.2$, $p < .01$; $\eta^2 = 0.049$), and even less in the PK acquisition condition ($M_{fake} = 3.76$, $SD = 1.91$; $M_{genuine} = 7.28$, $SD = 1.58$; $F(1, 392) = 154.9$, $p < .01$; $\eta^2 = 0.284$). In the control condition, the difference was marginally significant ($M_{fake} = 5.55$, $SD = 1.63$; $M_{genuine} = 6.18$, $SD = 1.50$; $F(1, 392) = 3.47$, $p = .06$; $\eta^2 = 0.009$). The larger differences in attributions of authenticity between genuine and fake reviews in the expected direction, PK acquisition > PK activation > Control, indicate that the manipulation was successful.

5.3.2. Trustworthiness

Tables 5 and 6 show that the results largely replicate those of Study 2

Table 5
ANOVA results (Study 3).

		F	η^2	M (SD)
(A) Persuasion Knowledge	Trustworthiness	2.49***	0.041	$M_{control}$ = 6.60 (1.37); $M_{PKactivation}$ = 6.14 (1.47); $M_{PKacquisition}$ = 6.16 (2.39)
	Review Favorability	10.5***	0.051	$M_{control}$ = 2.20 (1.40); $M_{PKactivation}$ = 2.93 (1.62); $M_{PKacquisition}$ = 3.09 (1.58)
	Purchase Intentions	22.3***	0.10	$M_{control}$ = 2.01 (1.35); $M_{PKactivation}$ = 3.08 (1.74); $M_{PKacquisition}$ = 3.30 (1.60)
(B) Review Authenticity	Trustworthiness	158***	0.29	M_{fake} = 5.34 (1.85); $M_{genuine}$ = 7.22 (1.27)
	Review Favorability	238***	0.037	M_{fake} = 1.81 (1.09); $M_{genuine}$ = 3.75 (1.41)
	Purchase Intentions	9.7**.	0.023	M_{fake} = 2.17 (1.37); $M_{genuine}$ = 3.54 (1.66)
(A) * (B)	Trustworthiness	29.4***	0.13	
	Review Favorability	.02$^{n.s.}$	<0.01	
	Purchase Intentions	4.6**	0.023	

Abbreviations: M = mean; SD = standard deviation; η^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

for trustworthiness. The main effects for both persuasion knowledge and review authenticity were significant (see Table 5). Moreover, trustworthiness toward the review was overall lower in the PK activation than in the control condition (p =.01). This reduction in trustworthiness occurred in both reviews, although only statistically significant in the fake review (mean difference fake = -0.61, p =.015; mean difference genuine = -0.40, p =.14). Overall, H6a was supported.

As predicted in H1, the persuasion knowledge × review authenticity interaction was significant. Pairwise comparisons revealed that the fake review was perceived as significantly less trustworthy than the genuine review in both the control and PK acquisition conditions, which supports H1b but not H1a. Moreover, the genuine review was perceived as more trustworthy than the fake review in the PK activation condition. As in Study 2, the results suggest that participants had some baseline persuasion knowledge about fake reviews. Nonetheless, like in Study 2, genuine vs fake differences in trustworthiness loom larger as one moves from control to PK activation to PK acquisition conditions, supporting the overarching hypothesis of the study that persuasion knowledge benefits consumers.

5.3.3. *Favorability*

The main effects of both persuasion knowledge and review authenticity were significant (see Table 5). Post hoc tests revealed that perceptions of the favorability of review were significantly higher in the PK activation condition compared to the control condition (p <.01), suggesting that warning made consumers discount the negativity of the fake review (i.e., judged them as more favorable). Further, this adjustment in favorability was observed for both reviews (mean difference fake = 0.66, p <.01; mean difference genuine = 0.60, p <.01), supporting H6. However, contrary to expectations, the interaction between persuasion knowledge and review authenticity was not significant, so H2 was not supported.

5.3.4. *Purchase intentions*

The main effects of both persuasion knowledge and review authenticity were significant (see Table 5). Post hoc tests revealed that purchase intentions were higher in the PK activation than in the control condition for both reviews (mean difference fake = 0.90, p <.01; mean difference genuine = 1.09, p <.01). As expected, activating participants' current knowledge made them suspicious in general, discounting the negativity of both reviews indiscriminately (H6c supported).

As predicted in H3, the interaction between persuasion knowledge and review authenticity was also significant. Post hoc tests revealed that purchase intentions of the genuine review vs fake review were higher in all conditions. Still, as expected, the difference was smaller in the PK acquisition than in the PK activation and control conditions (see Table 6).

The pattern of results for information search closely replicates those for purchase intentions. The more extreme an opinion, the less there should be a need to search for more information, provided the opinion is perceived as truthful. Fake reviews are more extreme and, as expected, participants searched for more information in the persuasion knowledge acquisition than in the others. Due to space limitations, we report the full results in Web Appendix D.

5.3.5. *Mediation analysis*

We conducted the same moderated mediation analysis with PK as a multicategorical variable as in Study 2. The direct effect was not moderated by PK, so we used Hayes' model 7. There was a positive and significant direct effect (b = 0.33, SE = 0.17, p =.04). The index of moderated mediation by trustworthiness was significant only for W2 (W1: −0.04 [-0.18: 0.09]; W2: −0.48 [-0.79: −0.22]), meaning that only the interaction between the mediating paths of the PK acquisition and control conditions was significantly different. The mediating path through trustworthiness was negative and significant for all conditions (control: -0.19 [-0.33: -0.08]; PK activation: -0.23 [-0.38: -0.11]; PK acquisition: −0.66 [-1.04: -0.35)]. Therefore, as in Study 2, H4 is partially supported, as we did not predict mediation by trust in the control condition (H4a). Indices of moderated mediation by review favorability were not significant (W1: −0.04 [-0.48: 0.41]; W2: -0.01 [-0.45: 0.42]), meaning that there was no difference between the mediation effects between the control and both PK conditions. Thus, H5 was not supported.

Table 6
Post hoc pairwise comparisons by persuasion knowledge acquisition conditions (Study 3).

	PK Acquisition				PK Activation				Control			
	Genuine	Fake	F	η^2	Genuine	Fake	F	η^2	Genuine	Fake	F	η^2
Trustworthiness	7.64	4.20	196***	0.33	6.75	5.57	23.2***	0.56	7.16	6.18	12.4***	0.031
	(1.27)	(2.09)			(1.20)	(1.47)			(1.14)	(1.39)		
Review Favorability	3.91	2.01	83.4***	0.18	3.89	2.03	79.9***	0.17	3.29	1.37	66.8***	0.15
	(1.36)	(1.15)			(1.43)	(1.24)			(1.41)	(0.61)		
Purchase Intentions	3.60	2.90	8.3***	0.021	3.95	2.26	48.6***	0.11	2.86	1.36	30.1***	0.071
	(1.57)	(1.55)			(1.72)	(1.31)			(1.54)	(0.67)		

Abbreviations: M = mean; SD = standard deviation; h^2 = partial eta squared
Significance levels: *p <.10; **p <.05; ***p <.01, n.s. - not significant (p >.10)

5.4. Discussion

Using negative fake reviews from a different service category, Study 3 supports our theoretical model but in the opposite direction. When consumers learned about fake review characteristics, they adjusted their judgments, reducing trustworthiness and, in the case of a negative review, discounted its negativity, which increased their purchase intentions. They also increased their willingness to search for more information about the service since they trusted less in the review. Unlike studies 1 and 2, however, consumers who acquired knowledge adjusted the favorability of the fake negative review to a lesser extent relative to the control condition. This is arguably due to negativity bias (i.e., a higher sensitivity to negative information than positive information; Baumeister et al., 2001), such that negative information more easily reverts an initial positive impression than positive information reverts an initial negative impression (Skowronski and Carlston, 1992).

6. Study 4

Studies 1–3 provided overall support for our theoretical model. However, contrary to our initial expectations, the results of Studies 2 and 3 suggest that participants in the control condition seemed to have some baseline persuasion knowledge, as they assigned lower levels of trustworthiness to the fake vs genuine review. Thus, the objective of this study was to examine whether consumers have some baseline knowledge of deception in online reviews and whether accuracy improves when consumers acquire persuasion knowledge.

6.1. Participants and design

Two hundred and nineteen individuals (M_{age} = 27.8; *SD* = 8.6; 51 % female) participated in a single-factor (persuasion knowledge: control vs acquisition) between-subjects online experiment. The participants were randomly assigned to one of the conditions. Attention check questions were included in the questionnaire, and the results were not significantly different when participants who failed at least one of these attention checks (N = 35) were excluded. Thus, our reports are based on the final sample of 219 respondents (see the results with the filtered sample in Web Appendix D). The procedure was pre-registered (https://aspredicted.org/RS1_CK6).

6.2. Procedure

Participants were informed that they would take part in a study about "people's judgments about online reviews." Next, they completed the online review persuasion knowledge scale with four items from Bambauser-Sachse and Mangold (2013) and the skepticism toward eWOM scale with five items taken from Zhang et al. (2016). As suggested by Ham et al. (2015), skepticism represents part of dispositional persuasion knowledge and is conceptually similar to persuasion knowledge. Both scales were measured by 9-point Likert-type questions (1 = Totally Disagree; 9 = Totally Agree). Importantly, we collected these measures at the beginning of the questionnaire so that persuasion knowledge manipulation would not influence the responses to these measures. The scale items and reliabilities are detailed in Web Appendix F.

Next, the participants were randomly assigned to either of the two persuasion knowledge acquisition experimental conditions, as in Study 1. We used the same detection guide for the PK manipulation as in studies 2 and 3, and the same neutral text for the "control" condition as in Study 3. After the manipulation of persuasion knowledge, participants were told that they would read ten reviews about hotels in Chicago, USA, taken from the internet and that "some of them were written by guests who stayed at the hotel, while others were artificially created." Unbeknownst to them, five reviews were fake, and five were genuine. For each review, they indicated whether the review was fake or genuine. The number of correct answers was the main dependent variable.

The five fake reviews were taken from Camilleri's (2019) quiz on fake review detection. The genuine reviews were taken from Yelp.com, following the same procedures as in previous studies, to ascertain that the review was truthful (see Web Appendix B). Additionally, we checked each review's authenticity through software designed by Ott et al. (2011), created specifically for hotels (https://www.reviewskpetic.com). The complete set of 10 reviews is shown in Web Appendix E. Importantly, we took the reviews as they were, without manipulating them, to allow for more external validity. Thus, compared to previous studies, the reviews in this study were more subtle and arguably harder to distinguish.

6.3. Results

The participant's overall persuasion knowledge about online reviews was mildly but significantly above the scale midpoint (=5) (M = 6.00, SD = 1.39, $t(218)$ = 10.6, p <.01). Similarly, participants were somewhat skeptical about eWOM (M = 6.27, SD = 1.48, $t(218)$ = 12.8, p <.01). There were no differences between conditions (all ps > 0.20). The results suggest that the participants possessed some baseline persuasion knowledge.

Next, we analyzed the participants' accuracy by checking their scores on the quiz. As shown in Fig. 2, the accuracy (i.e., the percentage of correct answers) was significantly higher in the PK condition (M_{PK} = 72.5 % vs $M_{control}$ = 61.7 %, χ^2 (1) = 28.5, p <.01). The results are more revealing when we analyze the participants' accuracies separately by review authenticity. Whereas the participants in the PK condition were considerably and significantly more accurate in detecting fake reviews when they acquired knowledge (M_{PK} = 66.5 % vs $M_{control}$ = 45.1 %, χ^2 (1) = 50.9, p <.01), there were no differences for genuine reviews (M_{PK} = 78.4 % vs $M_{control}$ = 78.3 %, χ^2 (1) = 0.00, p =.99, see Fig. 2). The higher accuracy in the set of genuine reviews is consistent with the truth bias found by Plotkina et al. (2020), as revealed by the higher number of reviews assigned as genuine (M = 6.14; SD = 1.51). This bias was much more pronounced in the control than in the PK condition (M_{PK} = 5.61, SD = 1.50 vs $M_{control}$ = 6.66, SD = 1.33, p <.01).

6.4. Discussion

Study 4 offered complementary evidence that consumers benefit from acquiring persuasion knowledge in online reviews. In addition, the results suggest that consumers have some baseline persuasion knowledge. However, the participants who were exposed to the fake review guide considerably improved their ability to discern a fake review from a genuine review.

7. Study 5

In studies 1–3, we combined the four fake review characteristics. But which of them has more discriminating power? To answer this question, we first conducted a pilot study to examine whether consumers perceive the 10 reviews of Study 4 as different in the four characteristics explained in the detection guide. Overall, there was a clear distinction between fake and genuine reviews concerning the four characteristics. Consistent with our assumptions, fake reviews were judged as highlighting more positive things, using more exaggerated language, using a promotional style, and describing things generically. Details of the results are provided in Appendix G. We then conducted another study to examine whether any of the four features of fake reviews have enough power to discriminate a genuine from a fake review, which we describe next.

7.1. Participants and design

Five hundred and two individuals (M_{age} = 32.6; SD = 12.2; 49 %

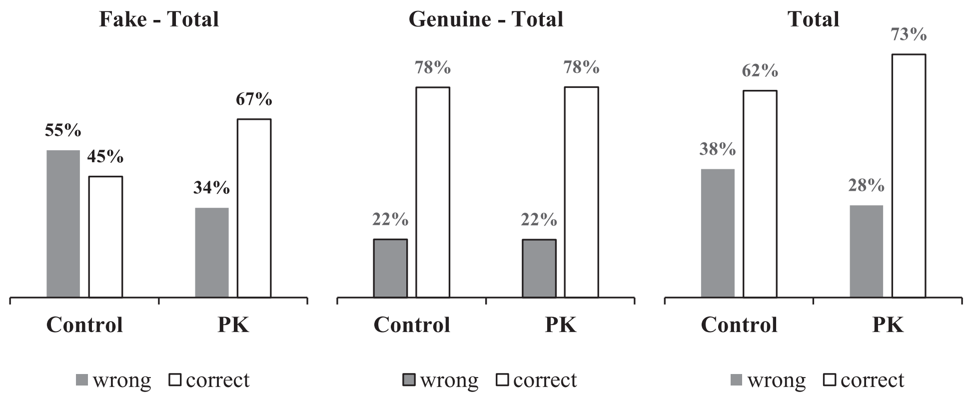

Fig. 2. A Comparison of the accuracy in the detection of fake, genuine, and total reviews between participants of the control and PK conditions (Study 4).

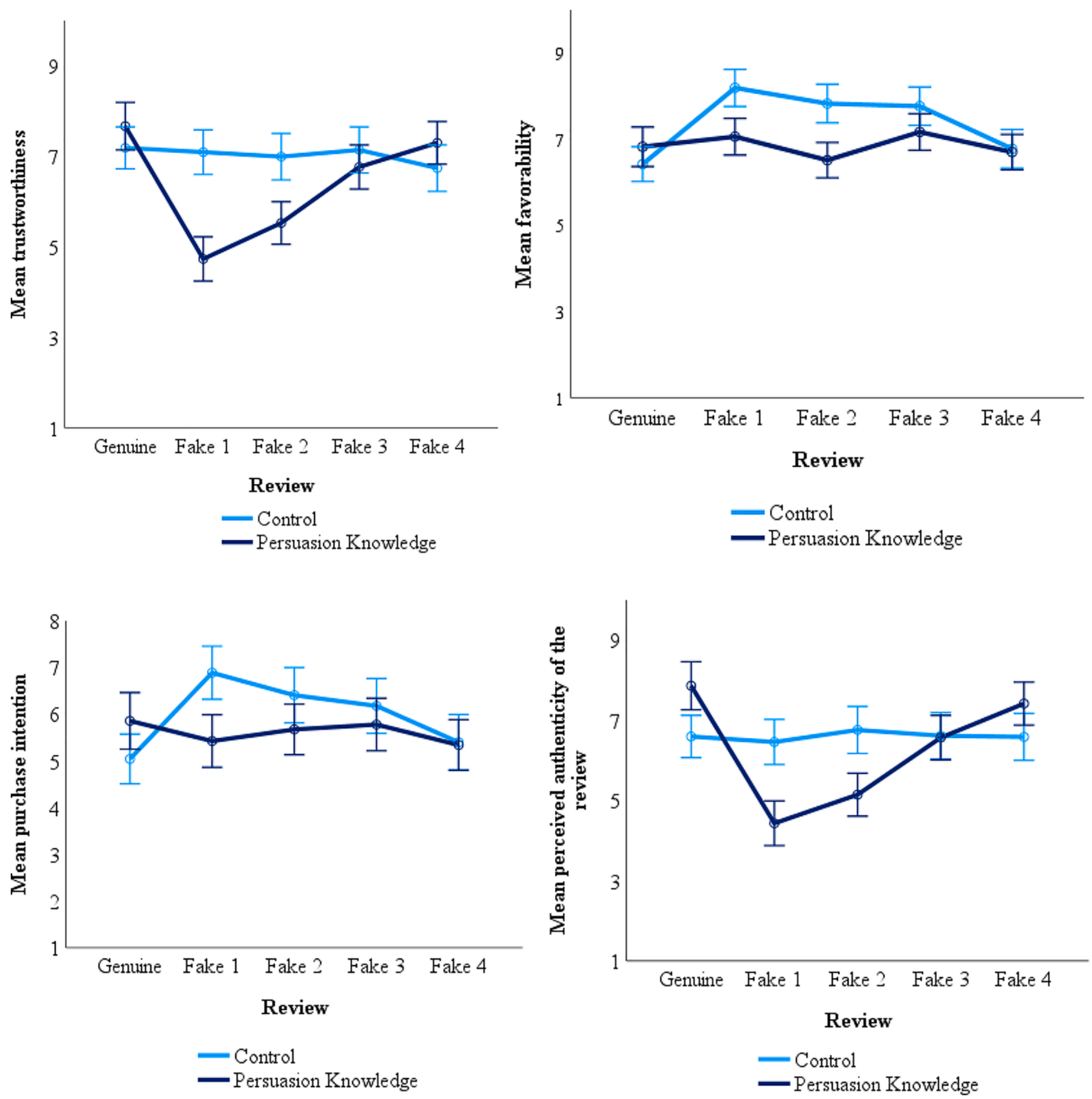

Fig. 3. Differences in trustworthiness, perceptions of favorability of the review, purchase intentions, and perceived authenticity of the review between a genuine and a fake review, for each type of fake review characteristic, between participants that acquired (Persuasion Knowledge) vs not acquired (Control) persuasion knowledge (Study 5). The error bars indicate the 95% confidence intervals.

female) participated in a 2 (persuasion knowledge: control vs acquisition) × 5 (review authenticity: genuine vs fake 1 vs fake 2 vs fake 3 vs fake 4) between-subjects design, online experiment. Participants were randomly assigned to conditions. Those who failed attention checks were automatically taken to the end of the survey and did not complete the study.

7.2. Procedure and measures

The procedure and measures in this study were the same as in Study 3, except that we had five review authenticity conditions. In this study, we created four fake reviews, each containing only one of the four features that discriminate genuine from fake reviews (fake 1: one-sidedness; fake 2: exaggeration and emotion; fake 3: personal selling style; fake 4: generic descriptions). Unlike previous studies, to create each of the four fake reviews, we only altered parts of the genuine review that were specific to the feature of each fake review. Details of the manipulations of each fake review relative to the genuine review are shown in Web Appendix E. The procedure was pre-registered (https://aspredicted.org/2M4_SSY).

7.3. Results

To compare each fake review with the genuine review, we regressed each dependent variable on review authenticity as a multicategorical predictor and persuasion knowledge (control = 0; acquisition = 1) as the moderator. We focused our analysis on the interaction effects for each dummy variable and the conditional effects at different values of the moderator. Due to space limitations, we present the details of the regressions, as well as means and standard deviations for each condition in the Web Appendix D. Across all the dependent measures, the results consistently reveal that the discriminating power of the fake review features is higher in the order fake1 > fake2 > fake3 > fake4. As shown in Fig. 3, review sidedness (fake1) was clearly the most discriminating feature, followed by exaggeration and emotions (fake2) and personal selling style (fake3). Descriptions of the experience (detailed vs generic; fake 4) did not predict differences between genuine and fake reviews.

7.3.1. Trustworthiness

There was a significant interaction of persuasion knowledge and review authenticity ($F(4, 492) = 12.7$; $p < .01$; $\eta^2 = 0.094$). The effect of review authenticity on trustworthiness was moderated by persuasion knowledge in all reviews except fake4. Analysis of conditional effects at values of the moderator indicates that, in the control condition, there were no differences in trustworthiness between the genuine and any of the four fake reviews. On the other hand, in the PK condition, the effects were negative and significant for all fake review features except fake 4. The magnitude of the absolute values of negative effects increases in the direction fake1 > fake2 > fake3 > fake4, meaning that participants trusted the fake review to a lesser extent, relative to the genuine, in this order.

7.3.2. Favorability

There was a significant interaction of persuasion knowledge and review authenticity ($F(4, 492) = 5.36$; $p < .01$; $\eta^2 = 0.042$). The regression results reveal an interaction effect in all reviews except fake 4. Analysis of conditional effects indicates that, in the control condition, the fake review was perceived as significantly more favorable than the genuine review, except for fake 4. The magnitude of the positive effect increases in the sequence fake1 > fake2 > fake 3 > fake4. On the other hand, in the PK condition, none of the effects were significant.

7.3.3. Purchase intentions

There was a significant interaction of persuasion knowledge and review authenticity ($F(4, 492) = 17.7$, $p < .01$, $\eta^2 = 0.033$). The regression results reveal an interaction effect in all reviews except for fake 4. Consistent with the previous studies, the analysis of conditional effects indicates that, in the control condition, the fake review rendered higher purchase intentions than the genuine review, except for fake 4. The magnitude of the positive effects increases in the sequence fake1 > fake2 > fake3 > fake4. On the other hand, in the PK condition, none of the effects were significant.

7.3.4. Perceptions of review authenticity

There was a significant interaction of persuasion knowledge and review authenticity ($F(4, 492) = 13.0$; $p < .01$; $\eta^2 = 0.096$). The regression results reveal an interaction effect between the genuine and all fake reviews, except fake4. In the control condition, there was no difference in perceptions of authenticity between a genuine and a fake review in any of the four fake review conditions. In contrast, in the PK condition, participants perceived the fake review as less authentic than the genuine review in all fake conditions, except for fake 4. The magnitude of the absolute values of the negative effects increases in the following sequence: fake1 > fake2 > fake3 > fake4.

7.4. Discussion

Study 5 examined whether any of the four characteristics that discriminate genuine against fake reviews used in combination in the previous studies has sufficient discriminatory power when they occur individually. The results are consistent across all dependent variables and suggest that the power increases in the direction fake1 > fake2 > fake3 > fake4. However, fake4 (describing the experience generically), although in the expected direction in all dependent measures, did not reach statistical significance. By far, sidedness is what most distinguished a fake from a genuine review. The two-sided genuine review, relative to the one-sided fake review, was significantly more trusted among those who acquired persuasion knowledge, but not among those in the control condition. In contrast, review favorability and purchase intentions were significantly higher for the one-sided fake, compared to the two-sided genuine review, among naive consumers, but not among those who acquired persuasion knowledge. These same patterns were also observed, although to a lesser extent, when the review contained more exaggeration and emotional content and, and when the spammer used a personal selling style when writing the review (e.g., "don't miss this opportunity").

8. General discussion

The present research investigated whether and how the acquisition of knowledge about features that distinguish fake from genuine reviews (i.e., online review persuasion knowledge) affects consumers' judgments when they read fake vs genuine reviews. Taken together, the results of our studies suggest that, although some baseline persuasion knowledge exists, fake reviews are much more likely to go unnoticed by consumers when they are not equipped with tools to detect them. As a result, they tend to trust fake and genuine reviews alike. Since spammers are compensated to promote (or denigrate) products by writing more extreme and one-sided fake reviews, they are perceived as more favorable (unfavorable) than genuine reviews, thus misleading consumers to higher (lower) purchase intentions. However, when consumers acquire some knowledge on how to detect them, they adjust their judgments accordingly. Their trust in the fake review is undermined, and they discount favorability (or unfavorability, in the case of negative reviews), thus reducing (or increasing, in the case of negative reviews) the purchase intentions of the target object of the review.

Our research contributes to the literature on fake reviews and persuasion knowledge on a few fronts. First, we show that persuasion knowledge acquisition in the domain of online reviews can aid consumers in distinguishing deceitful from truthful reviews to a larger extent compared to when they lack knowledge or when they are just alerted about fake reviews, as past research did (Bambauer-Sachse and

Mangold, 2013; Munzel, 2015, 2016). Moreover, we provide a methodological contribution since we manipulated persuasion knowledge to offer more firm causal inferences than past correlational studies (e.g., Bambauer-Sachse and Mangold, 2013; Zhuang et al., 2018). In addition, our effects occurred above and beyond the expertise in online reviews.

Second, we show that humans can detect fraudulent reviews, provided that the appropriate cues are supplied. Past research has consistently claimed that the accuracy of humans in detecting fake reviews is no higher than chance (Hovy, 2016; Ott et al., 2011; Plotkina et al., 2020; Salminen et al., 2022). This research stream compares the accuracy of detection algorithms developed for computers and humans. However, such algorithms are generally based on features that are less suited to humans, such as readability indices (e.g., Plotkina et al., 2020). By selecting fake review features from the literature that are more diagnostic for sifting fake from truthful reviews, we have shown that consumers can discern them.

Third, we contribute to the persuasion knowledge literature more broadly. Recently, Isaac and Grayson (2017, 2020) called attention to the fact that persuasion knowledge activation does not always lead to more skepticism and resistance to persuasion attempts – as many past studies have claimed. They showed that higher levels of persuasion knowledge help consumers distinguish between guileful and honest tactics. Likewise, the present research shows that consumers who acquire persuasion knowledge punish the guileful but not the genuine reviews. We go beyond and demonstrate that persuasion knowledge activation, by, for instance, priming the consumer with persuasion agents' ulterior motive, may not be as effective if the consumer lacks the knowledge per se. As other scholars have claimed, in specific contexts, such as children's exposure to advertising, interventions are necessary to develop persuasion knowledge and protect consumers (Austin and Johnson, 1997; Nelson, 2016; Rozendaal et al., 2011). The current research shows that online reviews are one such context.

Fourth, contrary to claims that humans are not able to detect fake reviews (Hovy, 2016; Ott et al., 2011; Plotkina et al., 2020; Salminen et al., 2022), the current research shows that people can detect deception when they learn some of the linguistic characteristics that are prevalent in fake reviews, at least when these characteristics are present.

Our research has important implications for practitioners as well. Recommendation and e-commerce websites should implement or enhance detection mechanisms in their platforms by developing fake review detection algorithms taking into account the features revealed in the current research. Also, because message-sidedness was shown to be highly diagnostic of fake reviews, managers could mitigate fake reviews by adopting a review system that obliges the user to comment on positive and negative points, as is currently done, for instance, on the Booking.com website. Also, consumers would benefit if they were aware of fake reviews and learned how to spot them. However, it is unlikely that managers of websites take this initiative since it can reduce the website's credibility (Steward et al., 2020). Thus, regulatory bodies could require websites to appropriately inform and protect consumers by imposing websites to adopt fake review detection and provide information on how to detect them.

Despite the robustness of our findings across the five experimental studies, we acknowledge some limitations. First, our studies used a genuine review and created a fake counterpart by changing parts of the text with features of fake review characteristics identified in past studies. In real-life settings, however, genuine reviews may contain some elements of fake reviews. In contrast, many fake reviews may not feature all the characteristics employed in the manipulations used in this study. For instance, a recent study by Banerjee (2022) showed that, although fake reviews are more emotive and extreme overall, in some cases, such as luxury hotels' positive reviews, fake and genuine reviews may be comparable in exaggeration. Additionally, the fake-genuine dichotomy may be blurred in many cases because some reviewers may have material rewards or gamification incentives. Therefore, although reviews may be based on real experiences, they might be inflated, as shown by Wu's (2019) study with Amazon's top reviewers. Thus, detecting a fake review despite the knowledge acquired may not be straightforward. As a result, some fake reviews may remain unnoticed, and consumers may become suspicious of truthful opinions. While the former is a typical Type I error (the null hypothesis that the review is fake is rejected when it is actually fake, i.e., a false positive), the latter is a Type II error (the null hypothesis that the review is fake is not rejected when it is actually not fake, i.e., a false negative). Because Type I and Type II errors are inversely related, persuasion knowledge acquisition may result in consumers discarding some honest opinions. However, as observed in our detection study, persuasion knowledge acquisition did not harm consumers' accuracy in detecting genuine reviews. However, we acknowledge that some genuine reviews can be overly positive and unbalanced, as is the case when consumers are delighted with the product or service. In such circumstances, these reviews can be wrongly taken as fake. The present study focused on reducing Type I errors, which may increase Type II errors, as per the previous discussion. This is a trade-off we all face in our daily lives. By protecting ourselves from scams, we may eventually risk passing on interesting opportunities. We believe, however, that the net balance is positive and that consumers are better off when they protect themselves from fraud, even if this comes at the expense of discarding a few genuine opinions of some overly satisfied peers. An interesting avenue for future studies would be examining the proportion of genuine reviews that have the characteristics of the fake reviews identified in this study. This could yield an estimation of the downside of persuasion knowledge acquisition.

Another interesting avenue for forthcoming studies would be examining whether fake detection algorithms sustain their detection capabilities over time. Banerjee and Chua (2014) have shown that fake review writers can easily mimic characteristics of authentic reviews. If this is a prevalent pattern, then spammers should learn to "beat the system," making it harder for both algorithms and humans to distinguish deception from truth. Finally, future studies could test other mechanisms of fake review detection. This study analyzed only textual review-centric features, but reviewer-centric features, such as those presented in Web Appendix A, could also be tested.

Moreover, website detection capabilities increase purchase intentions when the site informs fraud detection mechanisms (Munzel, 2015). Recently, software for fake review detection has been made available to the general public (e.g., fakespot.com, reviewmeta.com). Whether and how consumers use them to improve their decisions is an interesting avenue of research.

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CRediT authorship contribution statement

Murilo Carrazedo Marques da Costa Filho: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Diego Nogueira Rafael:** Methodology, Investigation. **Lucia Salmonson Guimarães Barros:** Writing – review & editing, Supervision, Formal analysis. **Eduardo Mesquita:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbusres.2022.113538.

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